COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
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HANDS-ON ACTIVITY NO. 1.1 Basic C++ Programming

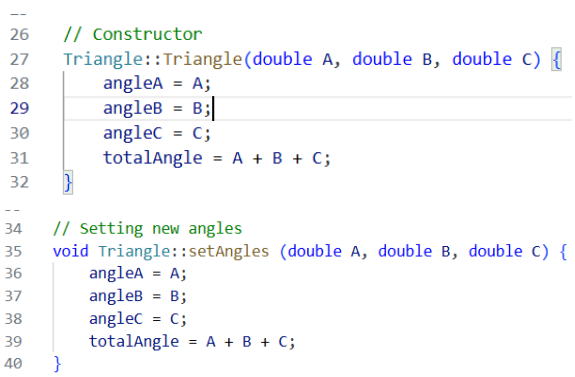
1. PROCEDURE OUTPUT

Today's hands-on activity tackled **Function Prototyping,** whose main purpose is to declare a function that tells the compiler/prgram to execute to be able to create and display the expected output, specifically the functions's name, return type,and parameters. With these, I could say that function prototyping is used to improve our codes and make it organized in a way where we can read it by separating the declaration and labeling it when implementing. To add, as I went through the end of the procedure activity, we also recalled the classes (like the public and private classes). Also, to add, we defined constructors where it automatically runs/executes when an object is created.

SECTIONS	ANSWERS
1. Header File Declaration Section	`#include <iostream>` - It is said to be the directives and imported libraries. - In C++, this line is a directive that calls the **INPUT / OUTPUT directory library.**
2. Global Declaration Section	- The **Global Declaration** holds the variables and constants of the code.

3. Class Declaration and Method Defining Section	`// function prototyping lesson`
	`class Triangle{`
	`private: // Private class`
	`  double totalAngle, angleA, angleB, angleC;`
	`public: // Public class`
	`// Constructor`
	`  Triangle (double A, double B, double C);`
	`  void setAngles (double A, double B, double C);`
	`  const bool validateTriangle();`
	`};`
	- Self-explanatory, this section is dedicated for the **class section** where data is declared.

4. Main Function	`int main(){`
	`  Triangle set1(50,90,75);`
	`  // Driver code`
	`  if (set1.validateTriangle()){`
	`  std::cout << "The shape is a valid triangle." << std::endl;}`
	`  else {`
	`  std::cout << "The shape is NOT a valid Triangle." << std::endl;}`
	`  return 0;`
	`}`

	- To be exact, the Ln 15 is the main function. - But, overall, the **main function** serves as where the execution of the program starts. Inside it, the code calls for the lines to be displayed like the: "The shape is a valid triangle."
5. Method Definition	 - In this section, it **contains the implementation of the code**. In my own understanding, this part is the **implementation because of the computation and analyzation** to be done of the code to be able to validate whether the triangle is truly a triangle or not.

```
26  // Constructor
27  Triangle::Triangle(double A, double B, double C) {
28      angleA = A;
29      angleB = B;
30      angleC = C;
31      totalAngle = A + B + C;
32  }

34  // Setting new angles
35  void Triangle::setAngles (double A, double B, double C) {
36      angleA = A;
37      angleB = B;
38      angleC = C;
39      totalAngle = A + B + C;
40  }
```

Table 1-1. C++ Structure Code for Answer

CODE:

HOA1.1.cpp

```cpp
#include <iostream>
// no using namespace std daw

// function prototyping lesson
class Triangle{
private: // Private class
    double totalAngle, angleA, angleB, angleC;
public: // Public class
// Constructor
    Triangle (double A, double B, double C);
    void setAngles (double A, double B, double C);
    const bool validateTriangle();
};

int main(){
    Triangle set1(100,30,50);
    // Driver code
    if (set1.validateTriangle()){
        std::cout << "The shape is a valid triangle." << std::endl;}
    else {
        std::cout << "The shape is NOT a valid Triangle." << std::endl;}

    return 0;
}
```

```
25
26  // Constructor
27  Triangle::Triangle(double A, double B, double C) {
28      angleA = A;
29      angleB = B;
30      angleC = C;
31      totalAngle = A + B + C;
32  }
33
34  // Setting new angles
35  void Triangle::setAngles (double A, double B, double C) {
36      angleA = A;
37      angleB = B;
38      angleC = C;
39      totalAngle = A + B + C;
40  }
41
42  // Computing the Triangle
43  const bool Triangle::validateTriangle() {
44      return (totalAngle == 180);}
```

OUTPUT:

```
15  int main(){
16      Triangle set1(100,30,50);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is a valid triangle.
```

```
15  int main(){
16      Triangle set1(40,30,110);
17      // Driver code
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is a valid triangle.
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is a valid triangle.
```

```
15  int main(){
16      Triangle set1(50,90,75);
17      // Driver code
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is a valid triangle.
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is a valid triangle.
PS C:\DSA> C:\DSA\HOA1.1.exe
The shape is NOT a valid Triangle.
PS C:\DSA>
```

Table 1-2. ILO Output Observation and Comments

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.

HOA1_SuppTask1.cpp > main()

```cpp
#include <iostream>
// still no using of namespace

void displaySum (
    int num1,
    int num2
);

int main(){
    displaySum(1345, 218);

    return 0;
}

void displaySum (
    int num1,
    int num2)
{
    std::cout << "The sum of the two numbers is: " << num1 + num2 << std::endl;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1_SuppTask1.exe
The sum of the two numbers is: 1563
```

- Create a function that will return whether variable A is greater than variable B.

HOA1_SuppTask2.cpp > main()

```cpp
#include <iostream>
// no using namespace

bool isGreater(int numA, int numB);

int main(){
    if (isGreater(89.5, 109)){
        std::cout << "A is GREATER than B." << std::endl;}
    else{
        std::cout << "A is NOT GREATER than B." << std::endl;}

    return 0;
}

bool isGreater(int numA, int numB){
    return (numA > numB);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1_SuppTask2.exe
A is NOT GREATER than B.
```

HOA1_SuppTask2.cpp > main()

```cpp
#include <iostream>
// no using namespace

bool isGreater(int numA, int numB);

int main(){
    if (isGreater(96, -1)){
        std::cout << "A is GREATER than B." << std::endl;}
    else{
        std::cout << "A is NOT GREATER than B." << std::endl;}

    return 0;
}

bool isGreater(int numA, int numB){
    return (numA > numB);
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\DSA> C:\DSA\HOA1_SuppTask2.exe
A is NOT GREATER than B.
PS C:\DSA> C:\DSA\HOA1_SuppTask2.exe
A is GREATER than B.
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

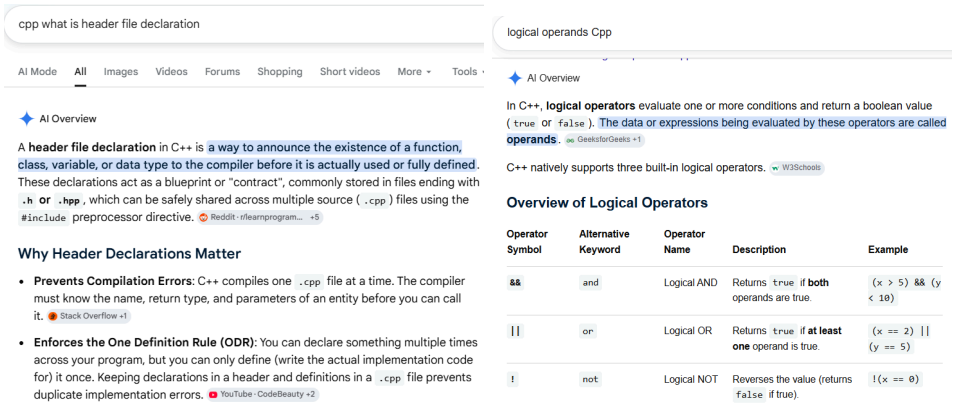
Tools Used: **Google AI Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Asking for clarifications, explanation of concepts.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:



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CONCLUSION:

- By the end of this Hands-On Activity, I got to refresh and gain knowledge in creating a program that has function prototyping wherein it is very useful in organizing the data code to be able to read, and analyze the it before implementing and executing it.
- Just as I addressed last time to my professor, I had a hard time putting the segments in their correct sequence, like which should go first. Although I admit, I'm still having a hard time comprehending it–and slow to work on, if I may add–I think I can at least work and progress on it bit by bit as I understand the main concepts and functions of it.